

Tugas SML: Customer Segmentation dengan Clustering

Dataset Online Retail II — UCI Machine Learning Repository

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1 Load Libraries

```
library(readxl)      # Baca file Excel
library(dplyr)        # Manipulasi data
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(lubridate)    # Parsing tanggal
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##   date, intersect, setdiff, union
```

```
library(ggplot2)      # Visualisasi
library(factoextra)    # fviz_nbclust, fviz_cluster
```

```
## Warning: package 'factoextra' was built under R version 4.5.3
```

```
## Welcome to factoextra!
```

```
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/products
```

```
library(cluster)      # silhouette()
```

```
## Warning: package 'cluster' was built under R version 4.5.3
```

```
library(mclust)        # GMM (Gaussian Mixture Model)
```

```
## Warning: package 'mclust' was built under R version 4.5.3
```

```
## Package 'mclust' version 6.1.2
```

```
## Type 'citation("mclust")' for citing this R package in publications.
```

```
##
```

```
## Attaching package: 'mclust'
```

```
## The following object is masked from 'package:dplyr':
##
##      count

library(fpc)          # cluster.stats() - Davies-Bouldin, Calinski-Harabasz

## Warning: package 'fpc' was built under R version 4.5.3

library(scales)       # comma() untuk format angka
library(clusterSim)   # index.DB() untuk Davies-Bouldin Index

## Warning: package 'clusterSim' was built under R version 4.5.3

## Loading required package: MASS

## Warning: package 'MASS' was built under R version 4.5.3

##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##      select

library(dbscan)       # DBSCAN dan kNNdistplot()

## Warning: package 'dbscan' was built under R version 4.5.3

##
## Attaching package: 'dbscan'

## The following object is masked from 'package:fpc':
##
##      dbscan

## The following object is masked from 'package:stats':
##
##      as.dendrogram
```

2 Import Dataset

Dataset **Online Retail II** dari UCI Machine Learning Repository berisi 541.909 transaksi toko online di Inggris tahun 2009–2011. Dataset tersedia dalam dua sheet Excel; kita gunakan kedua sheet dan gabungkan.

Download: <https://archive.uci.edu/dataset/502/online+retail+ii>

```
sheet1 <- read_excel("online_retail_II.xlsx", sheet = "Year 2009-2010")
sheet2 <- read_excel("online_retail_II.xlsx", sheet = "Year 2010-2011")
retail <- bind_rows(sheet1, sheet2)
```

```
cat("Dimensi awal:", nrow(retail), "baris x", ncol(retail), "kolom\n")
```

```
## Dimensi awal: 1067371 baris x 8 kolom
```

```
glimpse(retail)
```

```
## Rows: 1,067,371
```

```
## Columns: 8
```

```
## $ Invoice      <chr> "489434", "489434", "489434", "489434", "489434", "48943~
## $ StockCode    <chr> "85048", "79323P", "79323W", "22041", "21232", "22064", ~
## $ Description  <chr> "15CM CHRISTMAS GLASS BALL 20 LIGHTS", "PINK CHERRY LIGH~
## $ Quantity     <dbl> 12, 12, 12, 48, 24, 24, 24, 10, 12, 12, 24, 12, 10, 18, ~
## $ InvoiceDate   <dtm> 2009-12-01 07:45:00, 2009-12-01 07:45:00, 2009-12-01 07~
## $ Price        <dbl> 6.95, 6.75, 6.75, 2.10, 1.25, 1.65, 1.25, 5.95, 2.55, 3.~
## $ `Customer ID` <dbl> 13085, 13085, 13085, 13085, 13085, 13085, 13085, 13085, ~
## $ Country      <chr> "United Kingdom", "United Kingdom", "United Kingdom", "U~
```

2.1 Pemeriksaan Missing Value

Sebelum data dibersihkan, dilakukan pemeriksaan nilai kosong (NA) pada setiap kolom. Bagian ini penting karena clustering tidak dapat bekerja dengan baik jika fitur numerik masih mengandung nilai hilang.

```
missing_summary <- data.frame(
  Kolom      = names(retail),
  Missing    = colSums(is.na(retail)),
  Persentase = round(colSums(is.na(retail)) / nrow(retail) * 100, 2)
)

knitr::kable(missing_summary,
  caption = "Ringkasan Missing Value per Kolom")
```

Table 1: Ringkasan Missing Value per Kolom

	Kolom	Missing	Persentase
Invoice	Invoice	0	0.00

	Kolom	Missing	Persentase
StockCode	StockCode	0	0.00
Description	Description	4382	0.41
Quantity	Quantity	0	0.00
InvoiceDate	InvoiceDate	0	0.00
Price	Price	0	0.00
Customer ID	Customer ID	243007	22.77
Country	Country	0	0.00

```
cat("Total missing value:", sum(is.na(retail)), "\n")
```

```
## Total missing value: 247389
```

```
cat("Missing Customer ID:", sum(is.na(retail$`Customer ID`)), "\n")
```

```
## Missing Customer ID: 243007
```

3 Data Cleaning

Transaksi yang tidak valid (retur, harga/kuantitas negatif, CustomerID kosong) harus dihapus sebelum analisis.

```
retail_clean <- retail |>
  # Hapus baris CustomerID kosong
  filter(!is.na(`Customer ID`)) |>
  # Hapus transaksi retur (Invoice diawali 'C')
  filter(!grepl("^C", Invoice)) |>
  # Pastikan Quantity dan Price positif
  filter(Quantity > 0, Price > 0)

cat("Setelah cleaning:", nrow(retail_clean), "baris\n")
```

```
## Setelah cleaning: 805549 baris
```

```
cat("Jumlah unique customer:", n_distinct(retail_clean$`Customer ID`), "\n")
```

```
## Jumlah unique customer: 5878
```

4 Pembentukan Fitur RFM

Data transaksi diubah menjadi **satu baris per pelanggan** menggunakan analisis **RFM** (Recency, Frequency, Monetary).

Fitur	Definisi
Recency	Hari sejak transaksi terakhir (lebih kecil = lebih aktif)
Frequency	Jumlah invoice unik (lebih besar = lebih loyal)
Monetary	Total nilai belanja dalam GBP

```
# Tanggal referensi = satu hari setelah transaksi terakhir
ref_date <- max(retail_clean$InvoiceDate, na.rm = TRUE) + days(1)

rfm <- retail_clean |>
  mutate(TotalLine = Quantity * Price) |>
  group_by(CustomerID = `Customer ID`) |>
  summarise(
    Recency = as.numeric(difftime(ref_date,
                                   max(InvoiceDate, na.rm = TRUE),
                                   units = "days")),
    Frequency = n_distinct(Invoice),
    Monetary = sum(TotalLine, na.rm = TRUE)
  ) |>
  ungroup()

cat("Jumlah baris RFM:", nrow(rfm), "\n")
```

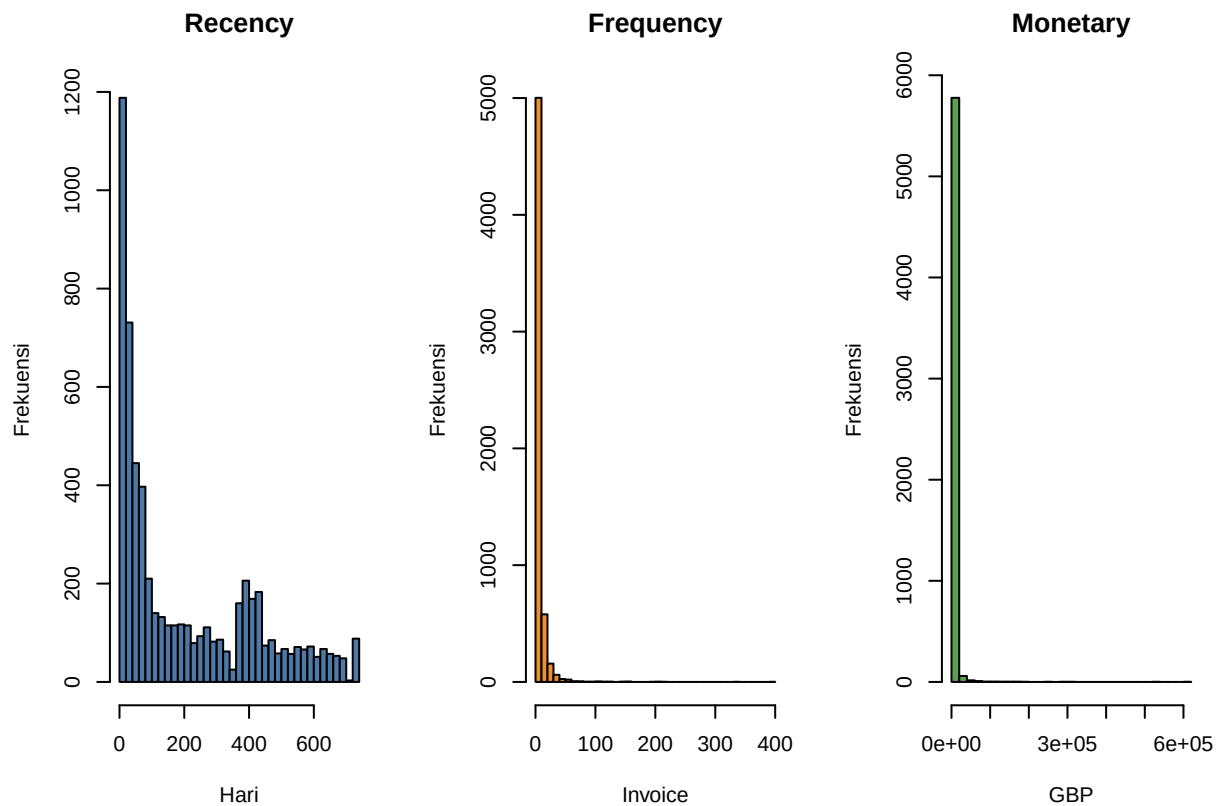
```
## Jumlah baris RFM: 5878
```

```
summary(rfm[, c("Recency", "Frequency", "Monetary")])
```

```
##      Recency      Frequency      Monetary
## Min.   : 1.00   Min.   : 1.000   Min.   : 2.95
## 1st Qu.: 26.05   1st Qu.: 1.000   1st Qu.: 348.76
## Median : 96.04   Median : 3.000   Median : 898.92
## Mean   :201.85   Mean   : 6.289   Mean   : 3018.62
## 3rd Qu.:380.10   3rd Qu.: 7.000   3rd Qu.: 2307.09
## Max.   :739.12   Max.   :398.000   Max.   :608821.65
```

4.1 Distribusi Fitur RFM

```
par(mfrow = c(1, 3))
hist(rfm$Recency, breaks = 40, col = "#4E79A7", main = "Recency",
     xlab = "Hari", ylab = "Frekuensi")
hist(rfm$Frequency, breaks = 40, col = "#F28E2B", main = "Frequency",
     xlab = "Invoice", ylab = "Frekuensi")
hist(rfm$Monetary, breaks = 40, col = "#59A14F", main = "Monetary",
     xlab = "GBP", ylab = "Frekuensi")
```



```
par(mfrow = c(1, 1))
```

5 Scaling

Karena nilai Monetary bisa ribuan kali lebih besar dari Recency, variabel perlu distandarisasi agar jarak antar-objek tidak didominasi satu fitur.

```
# Ganti bagian scaling dengan ini:
rfm_log <- rfm |>
  dplyr::select(Recency, Frequency, Monetary) |>
```

```
mutate(
  Recency    = log1p(Recency),
  Frequency  = log1p(Frequency),
  Monetary   = log1p(Monetary)
)

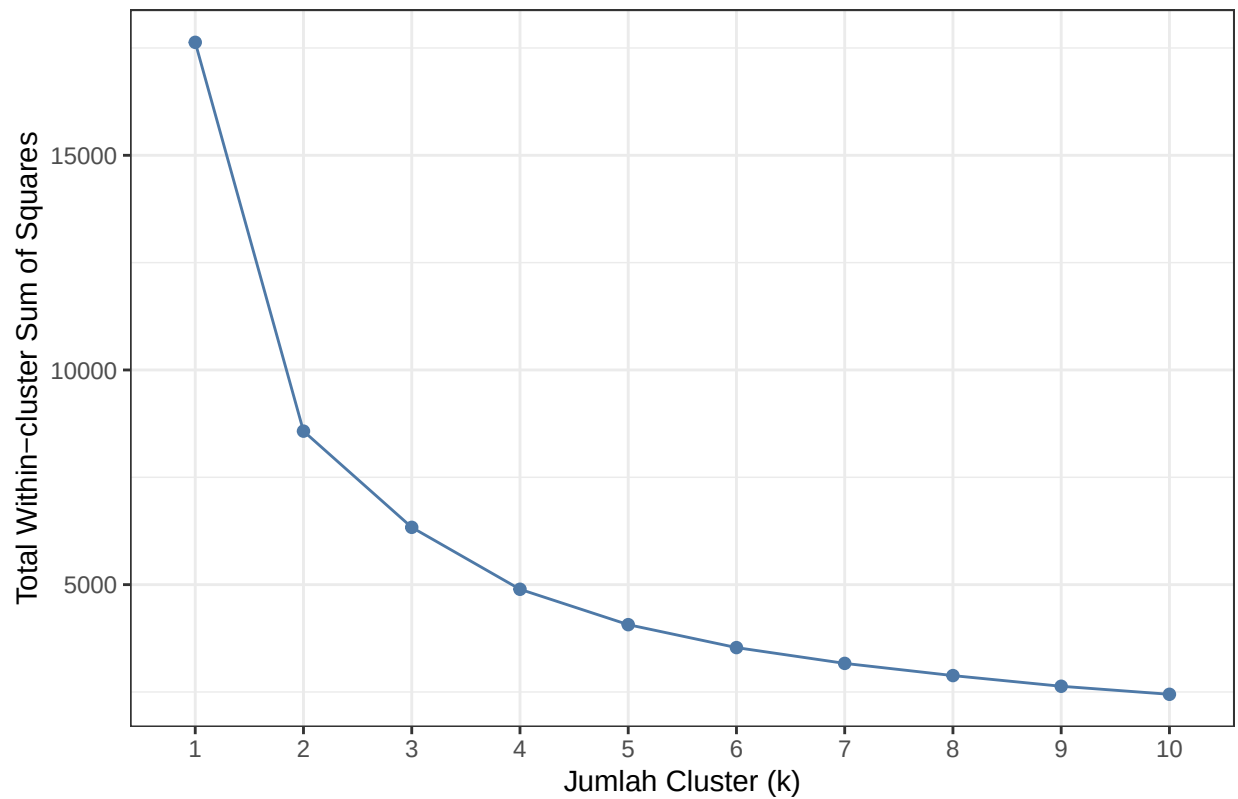
rfm_scaled <- scale(rfm_log) |> as.data.frame()
```

6 Penentuan Jumlah Cluster Optimal

6.1 Elbow Method

```
set.seed(16)
fviz_nbclust(rfm_scaled, kmeans,
  method = "wss",
  k.max   = 10,
  linecolor = "#4E79A7") +
labs(title = "Elbow Method - Menentukan k Optimal",
  x       = "Jumlah Cluster (k)",
  y       = "Total Within-cluster Sum of Squares") +
theme_bw()
```

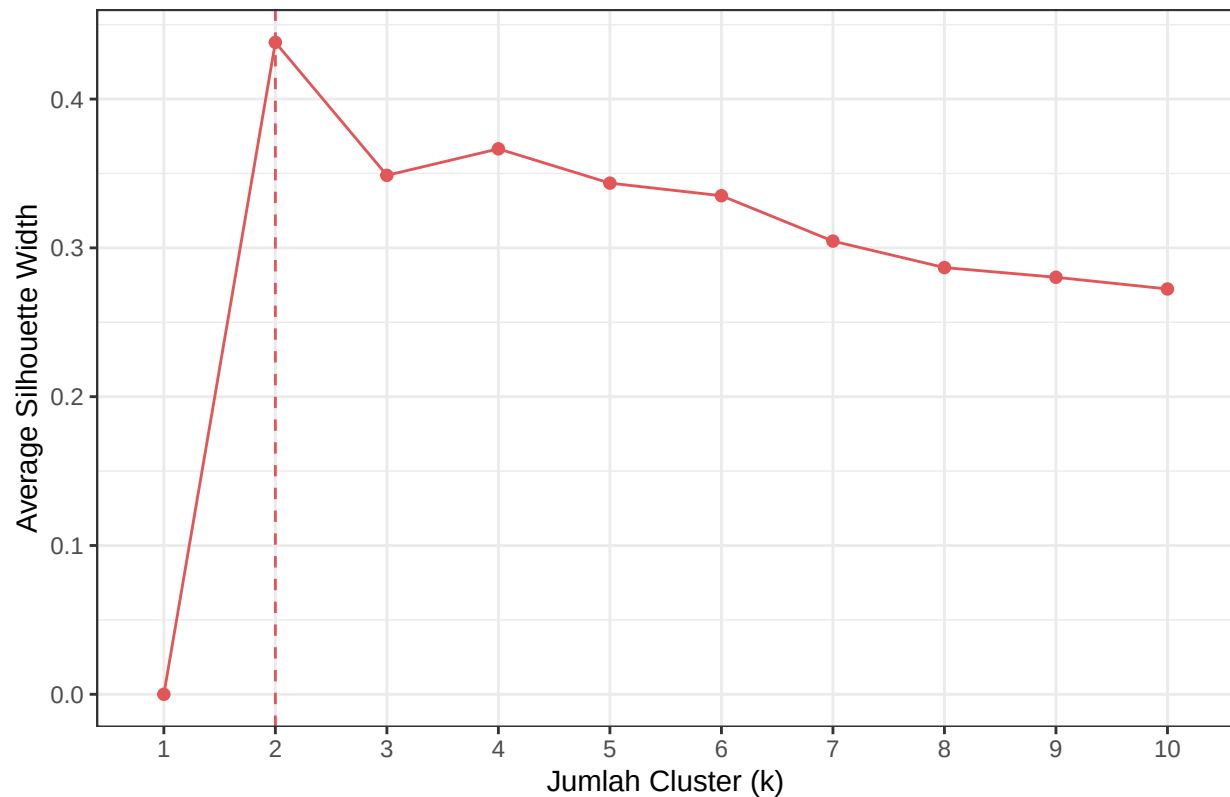
Elbow Method — Menentukan k Optimal



6.2 Silhouette Method

```
set.seed(16)
fviz_nbclust(rfm_scaled, kmeans,
             method = "silhouette",
             k.max = 10,
             linecolor = "#E15759") +
  labs(title = "Silhouette Method - Menentukan k Optimal",
       x = "Jumlah Cluster (k)",
       y = "Average Silhouette Width") +
  theme_bw()
```

Silhouette Method — Menentukan k Optimal



Kesimpulan: Berdasarkan plot Silhouette, jumlah cluster optimal yang dipilih adalah $k = 2$. Nilai Silhouette tertinggi berada pada $k = 2$, sehingga pemisahan cluster paling baik secara statistik.

```
k_opt <- 2
```

7 Metode Clustering 1: K-Means

K-Means adalah algoritma partisi yang meminimalkan jumlah kuadrat jarak setiap titik ke centroid klusternya.

```
set.seed(16)
km_result <- kmeans(rfm_scaled, centers = k_opt, nstart = 25, iter.max = 300)

rfm$Cluster_KMeans <- as.factor(km_result$cluster)
cat("Ukuran tiap cluster K-Means:\n")

## Ukuran tiap cluster K-Means:
```

```
print(table(rfm$Cluster_KMeans))
```

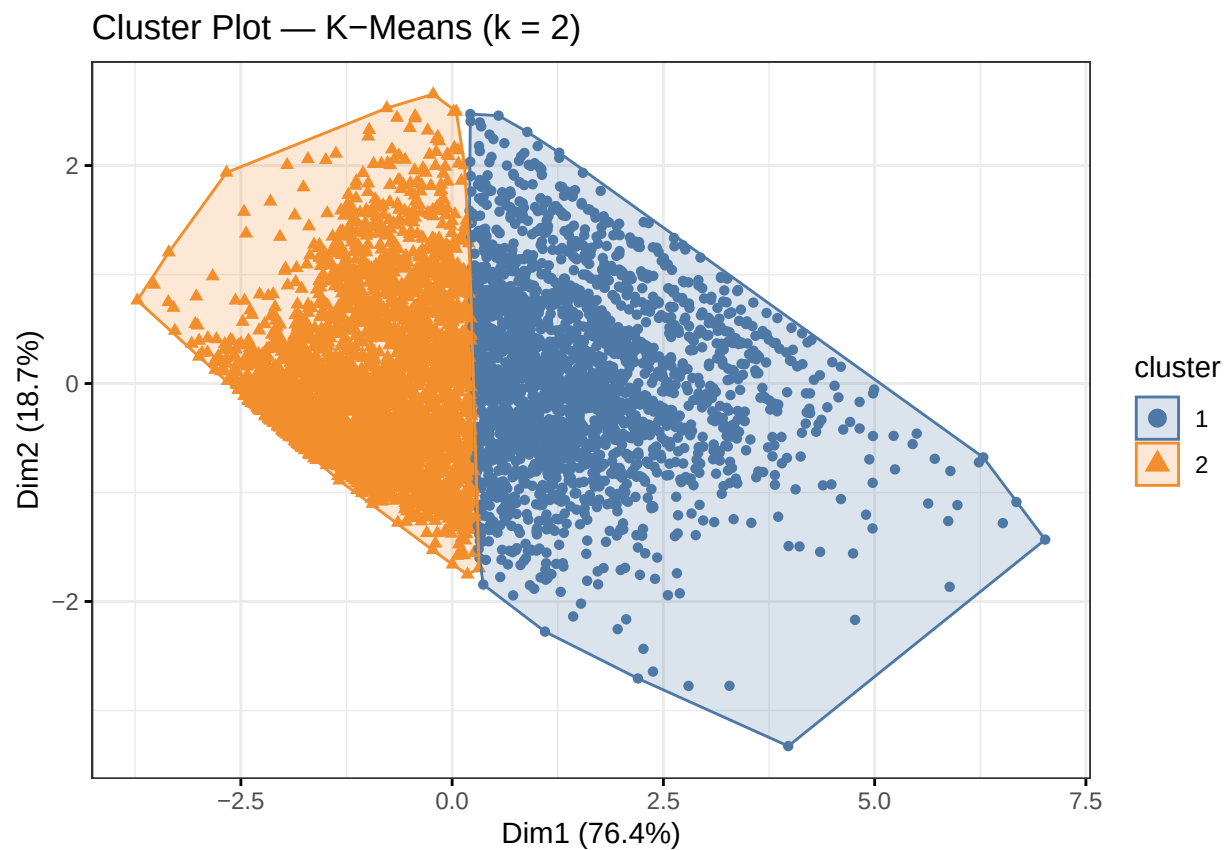
```
##
```

```
##      1      2
```

```
## 2357 3521
```

7.1 Visualisasi K-Means (PCA 2D)

```
fviz_cluster(km_result, data = rfm_scaled,  
             geom          = "point",  
             ellipse.type= "convex",  
             palette       = c("#4E79A7", "#F28E2B"),  
             ggtheme       = theme_bw(),  
             main          = "Cluster Plot - K-Means (k = 2)" +  
             theme(legend.position = "right")
```



8 Metode Clustering 2: Gaussian Mixture Model (GMM)

GMM adalah pendekatan berbasis model yang mengasumsikan setiap cluster mengikuti distribusi Gaussian. Berbeda dengan K-Means, GMM memberikan **probabilitas keanggotaan** (soft assignment) dan mampu menangkap bentuk cluster yang lebih fleksibel melalui matriks kovarians.

```
set.seed(16)
gmm_result <- Mclust(rfm_scaled, G = k_opt, verbose = FALSE)

rfm$Cluster_GMM <- as.factor(gmm_result$classification)
cat("Model GMM terpilih:", gmm_result$modelName, "\n")

## Model GMM terpilih: VVV

cat("Ukuran tiap cluster GMM:\n")

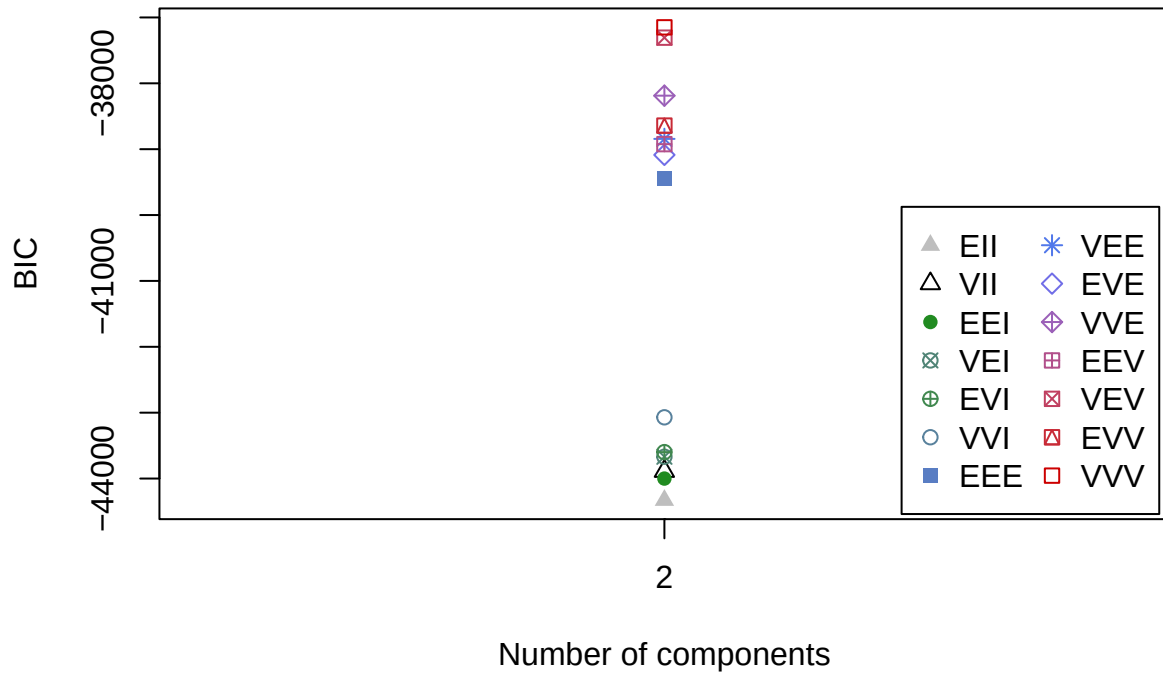
## Ukuran tiap cluster GMM:

print(table(rfm$Cluster_GMM))

##
##      1      2
## 1794 4084
```

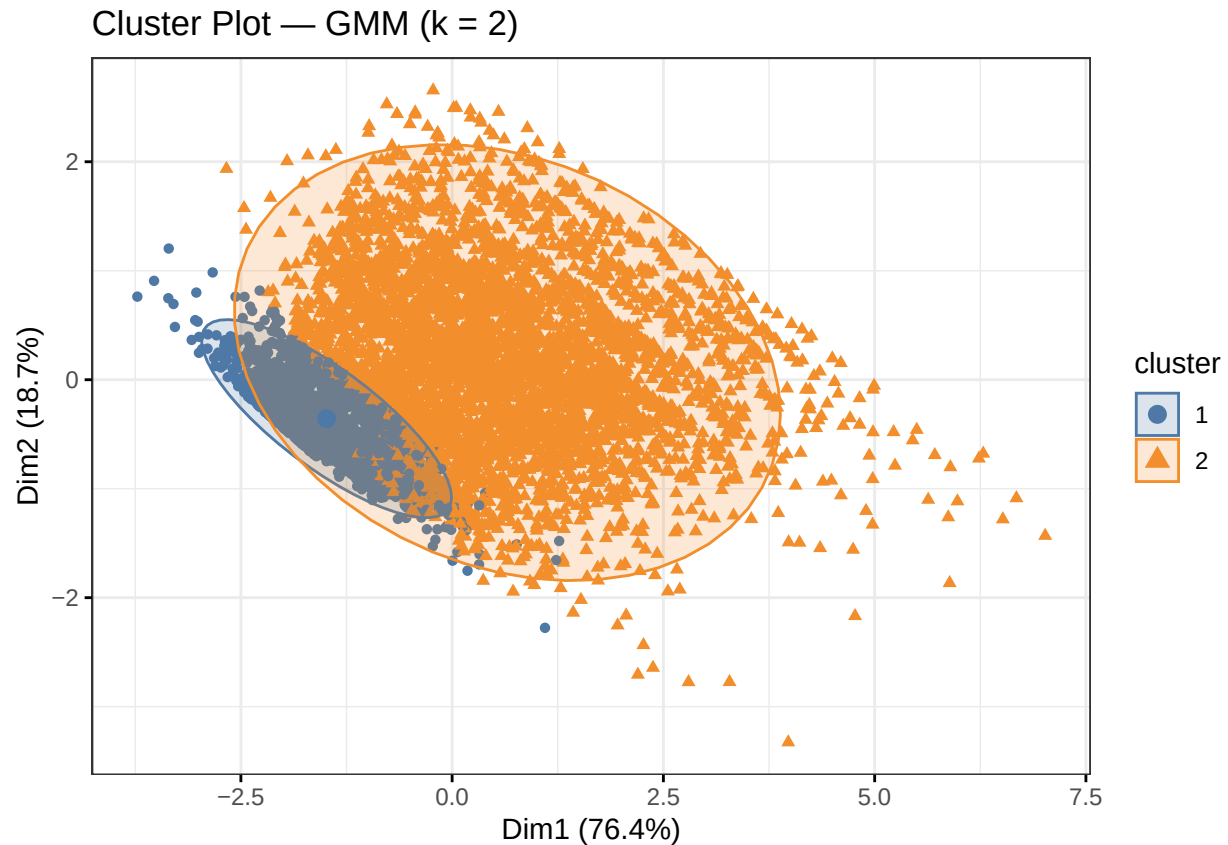
8.1 Visualisasi GMM (BIC)

```
plot(gmm_result, what = "BIC",
     main = "BIC per Model GMM")
```



8.2 Visualisasi GMM (Cluster Plot PCA 2D)

```
fviz_cluster(list(data = rfm_scaled,
                  cluster = gmm_result$classification),
             geom = "point",
             ellipse.type = "norm",
             palette = c("#4E79A7", "#F28E2B"),
             ggtheme = theme_bw(),
             main = "Cluster Plot - GMM (k = 2)" +
             theme(legend.position = "right"))
```



9 Metode Clustering 3: Hierarchical Clustering

Hierarchical Clustering digunakan untuk membentuk struktur kluster bertingkat. Pada analisis ini digunakan **Average Linkage** (`method = "average"`). Jumlah cluster ditentukan dengan memotong dendrogram menggunakan `cutree()` berdasarkan nilai `k_opt`.

```
dist_rfm <- dist(rfm_scaled, method = "euclidean")

hc_average <- hclust(dist_rfm, method = "average")
rfm$Cluster_HClust <- as.factor(cutree(hc_average, k = k_opt))

cat("Ukuran tiap cluster Hierarchical Average Linkage:\n")
```

```
## Ukuran tiap cluster Hierarchical Average Linkage:
```

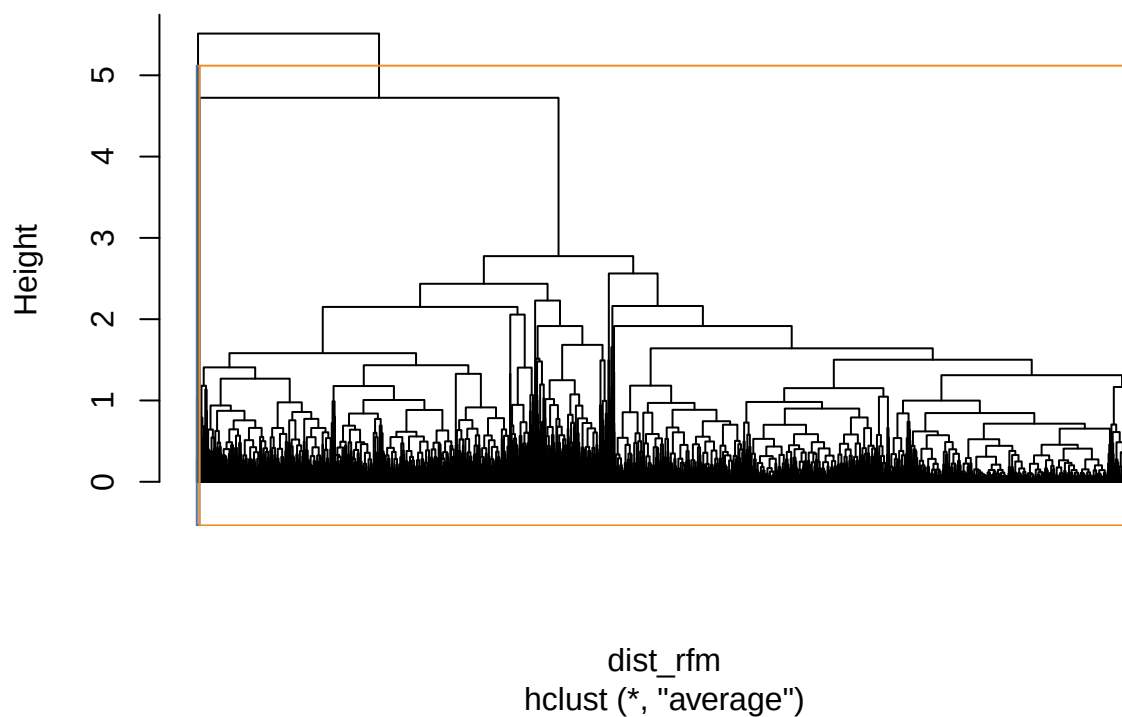
```
print(table(rfm$Cluster_HClust))
```

```
##
##      1      2
## 17 5861
```

9.1 Visualisasi Dendrogram dan Cutting Tree

```
plot(hc_average,  
     labels = FALSE,  
     hang   = -1,  
     main   = "Dendrogram - Hierarchical Clustering Average Linkage")  
rect.hclust(hc_average, k = k_opt, border = c("#4E79A7", "#F28E2B"))
```

Dendrogram — Hierarchical Clustering Average Linkage



10 Metode Clustering 4: Partitioning Around Medoids (K-Medoids)

PAM atau K-Medoids mirip dengan K-Means, tetapi pusat cluster menggunakan objek aktual (*medoid*) sehingga lebih tahan terhadap outlier.

```
set.seed(16)  
pam_result <- pam(rfm_scaled, k = k_opt)  
  
rfm$Cluster_PAM <- as.factor(pam_result$clustering)  
cat("Ukuran tiap cluster PAM / K-Medoids:\n")
```

```
## Ukuran tiap cluster PAM / K-Medoids:
```

```
print(table(rfm$Cluster_PAM))
```

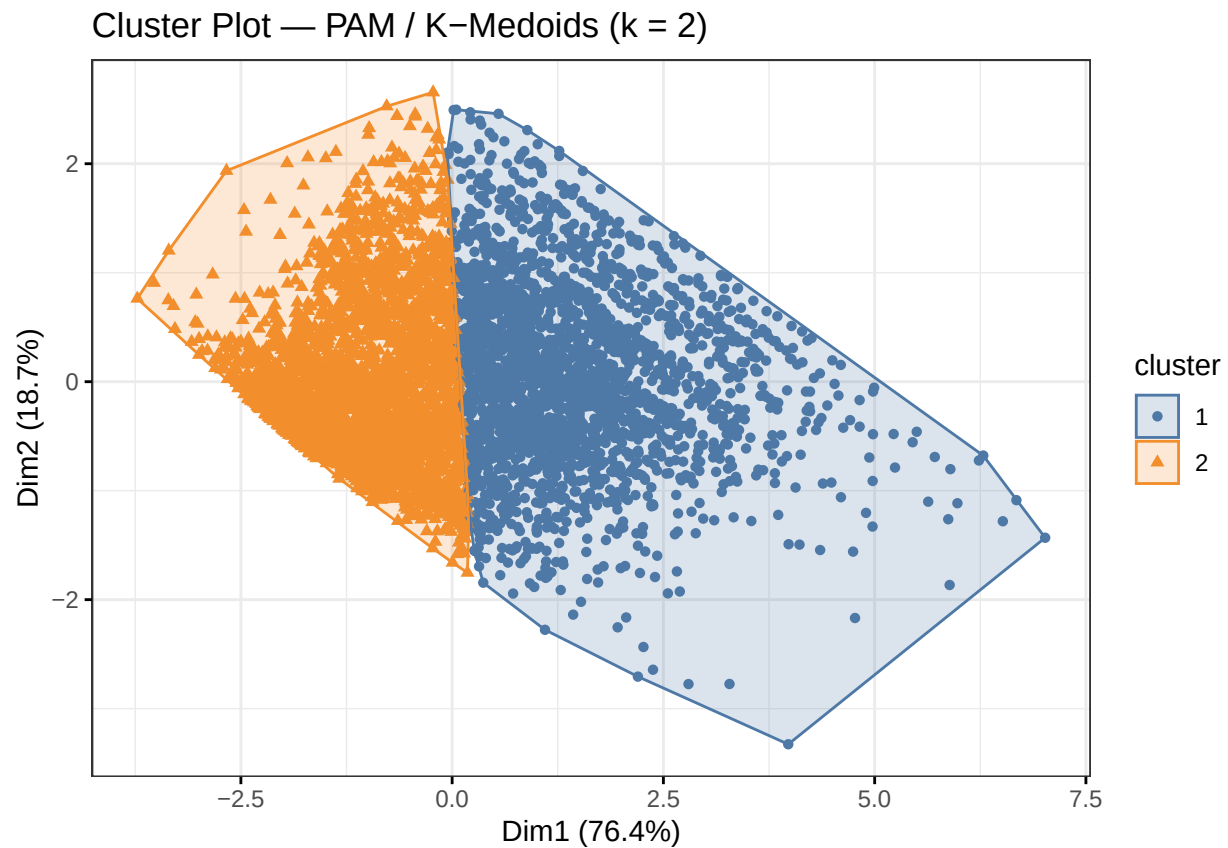
```
##
```

```
##      1      2
```

```
## 2530 3348
```

10.1 Visualisasi PAM (PCA 2D)

```
fviz_cluster(pam_result,  
             data      = rfm_scaled,  
             geom      = "point",  
             palette    = c("#4E79A7", "#F28E2B"),  
             ggtheme    = theme_bw(),  
             main       = "Cluster Plot - PAM / K-Medoids (k = 2)" +  
             theme(legend.position = "right")
```



11 Metode Clustering 5: DBSCAN

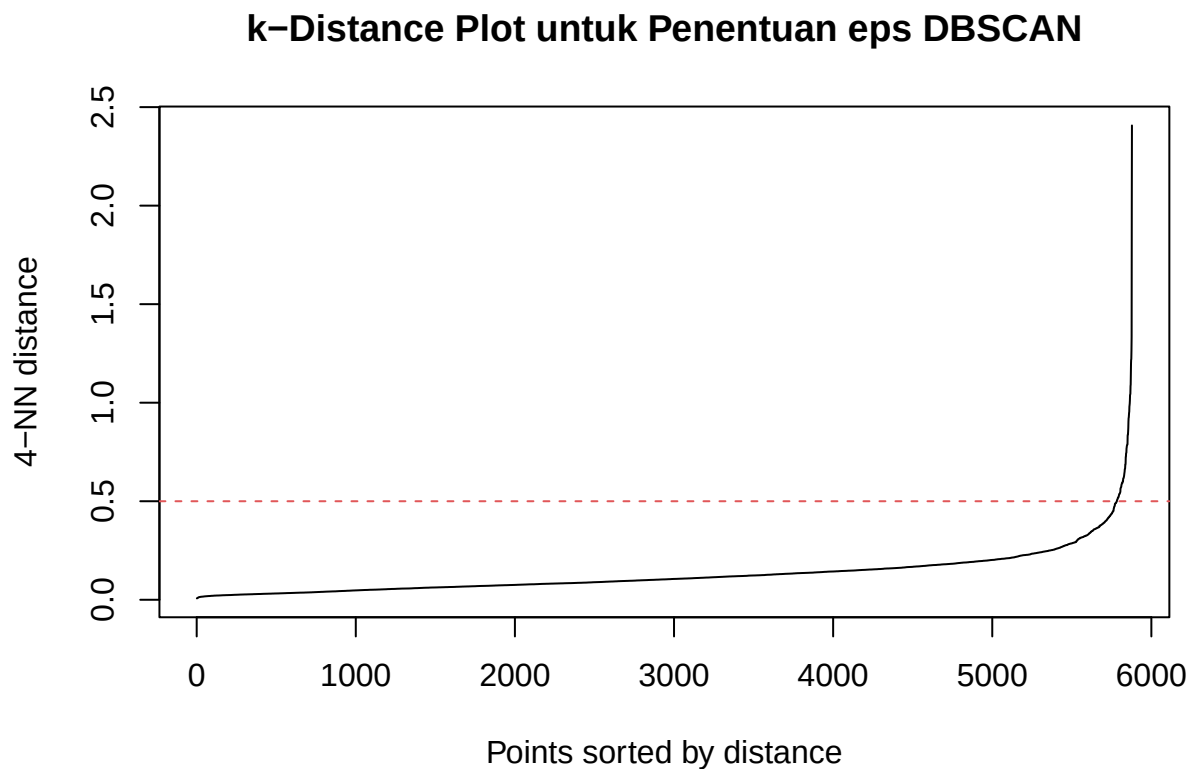
DBSCAN mengelompokkan data berdasarkan kepadatan titik dan dapat mendeteksi noise/outlier. Parameter utama yang perlu ditentukan adalah `eps` dan `minPts`.

11.1 Penentuan `eps` dengan k-Distance Plot

Plot k-distance digunakan untuk mencari titik siku (*elbow*) sebagai kandidat nilai `eps`. Nilai `eps_opt` di bawah dapat disesuaikan berdasarkan bentuk plot.

```
min_pts <- 4

kNNdistplot(rfm_scaled, k = min_pts)
abline(h = 0.5, lty = 2, col = "#E15759")
title("k-Distance Plot untuk Penentuan eps DBSCAN")
```



```
eps_opt <- 0.5

dbscan_result <- dbscan::dbscan(rfm_scaled, eps = eps_opt, minPts = min_pts)

rfm$Cluster_DBSCAN <- as.factor(dbscan_result$cluster)
```

```
cat("Ukuran tiap cluster DBSCAN, 0 = noise:\n")
```

```
## Ukuran tiap cluster DBSCAN, 0 = noise:
```

```
print(table(rfm$Cluster_DBSCAN))
```

```
##
```

```
##    0    1    2
```

```
##  59 4203 1616
```

12 Validasi Cluster

Validasi menggunakan kriteria **internal** karena tidak ada label kelas yang diketahui (unsupervised).

12.1 Silhouette Width

Mengukur seberapa mirip setiap objek dengan klusternya sendiri dibanding kluster lain. Nilai mendekati **1 = baik**.

```
dist_rfm <- dist(rfm_scaled)
```

```
sil_km <- silhouette(km_result$cluster, dist_rfm)
```

```
sil_gmm <- silhouette(gmm_result$classification, dist_rfm)
```

```
avg_sil_km <- mean(sil_km[, "sil_width"])
```

```
avg_sil_gmm <- mean(sil_gmm[, "sil_width"])
```

```
cat("Rata-rata Silhouette K-Means :", round(avg_sil_km, 4), "\n")
```

```
## Rata-rata Silhouette K-Means : 0.4381
```

```
cat("Rata-rata Silhouette GMM      :", round(avg_sil_gmm, 4), "\n")
```

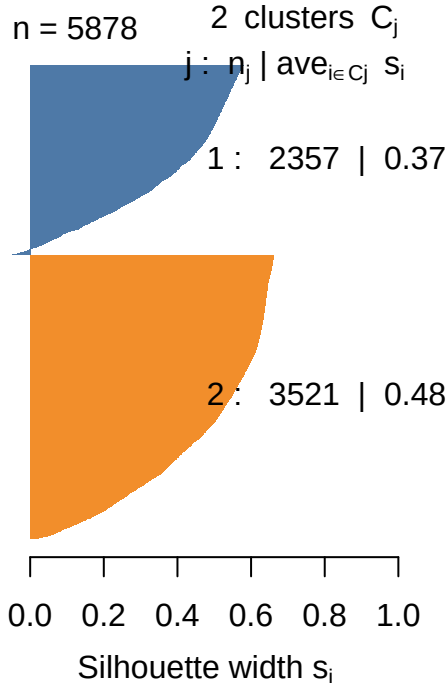
```
## Rata-rata Silhouette GMM      : 0.2924
```

```
par(mfrow = c(1, 2))
```

```
plot(sil_km, col = c("#4E79A7", "#F28E2B"),  
     border = NA, main = "Silhouette - K-Means")
```

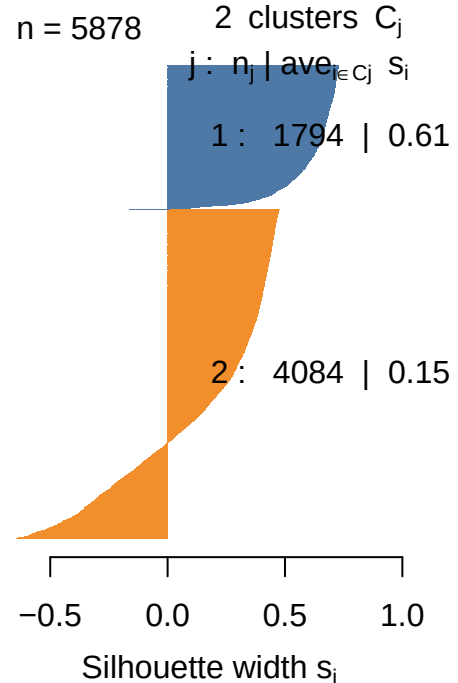
```
plot(sil_gmm, col = c("#4E79A7", "#F28E2B"),  
     border = NA, main = "Silhouette - GMM")
```

Silhouette — K-Means



Average silhouette width : 0.44

Silhouette — GMM



Average silhouette width : 0.29

```
par(mfrow = c(1, 1))
```

12.2 Davies-Bouldin Index

Mengukur rata-rata rasio dispersi dalam kluster terhadap jarak antar kluster. Nilai **lebih kecil** = **lebih baik**.

```
db_km <- index.DB(rfm_scaled, km_result$cluster)$DB
db_gmm <- index.DB(rfm_scaled, gmm_result$classification)$DB
```

```
cat("Davies-Bouldin K-Means :", round(db_km, 4), "\n")
```

```
## Davies-Bouldin K-Means : 0.9693
```

```
cat("Davies-Bouldin GMM      :", round(db_gmm, 4), "\n")
```

```
## Davies-Bouldin GMM      : 1.0982
```

12.3 Calinski-Harabasz Index

Rasio varians antar-kluster terhadap varians dalam-kluster. Nilai **lebih besar** = **lebih baik**.

```
stats_km <- cluster.stats(dist_rfm, km_result$cluster)
stats_gmm <- cluster.stats(dist_rfm, gmm_result$classification)

cat("Calinski-Harabasz K-Means :", round(stats_km$ch, 2), "\n")

## Calinski-Harabasz K-Means : 6205.27

cat("Calinski-Harabasz GMM      :", round(stats_gmm$ch, 2), "\n")

## Calinski-Harabasz GMM      : 3056.2
```

12.4 Validasi Lengkap Semua Algoritma

Bagian ini memperluas validasi internal agar mencakup K-Means, GMM, Hierarchical Clustering, PAM/K-Medoids, dan DBSCAN. Untuk DBSCAN, noise (`cluster = 0`) dikeluarkan dari perhitungan metrik internal karena noise tidak dihitung sebagai cluster evaluasi.

```
validate_cluster <- function(cluster_label, method_name, exclude_noise = FALSE) {
  label <- as.character(cluster_label)
  keep <- !is.na(label)

  if (exclude_noise) {
    keep <- keep & label != "0"
  }

  label_keep <- label[keep]

  if (length(unique(label_keep)) < 2 || sum(keep) < 3) {
    return(data.frame(
      Metode          = method_name,
      N               = sum(keep),
      Jumlah.Cluster  = length(unique(label_keep)),
      Noise           = sum(label == "0", na.rm = TRUE),
      Silhouette      = NA,
      Shadow.Value    = NA,
      Davies.Bouldin  = NA,
      Calinski.Harabasz = NA
    ))
  }
}
```

```

x <- as.data.frame(rfm_scaled[keep, , drop = FALSE])
y <- as.integer(as.factor(label_keep))
d <- dist(x)

sil <- mean(silhouette(y, d)[, 3])
st <- cluster.stats(d, y)
db <- tryCatch(index.DB(x, y)$DB, error = function(e) NA)

# Beberapa versi cluster.stats tidak menyediakan field shadow eksplisit.
shadow_value <- if ("shadow" %in% names(st)) {
  st$shadow
} else if ("avg.silwidth" %in% names(st)) {
  st$avg.silwidth
} else {
  NA
}

data.frame(
  Metode = method_name,
  N = sum(keep),
  Jumlah.Cluster = length(unique(label_keep)),
  Noise = sum(label == "0", na.rm = TRUE),
  Silhouette = round(sil, 4),
  Shadow.Value = round(shadow_value, 4),
  Davies.Bouldin = round(db, 4),
  Calinski.Harabasz = round(st$ch, 2)
)
}

validasi_semua <- bind_rows(
  validate_cluster(rfm$Cluster_KMeans, "K-Means"),
  validate_cluster(rfm$Cluster_GMM, "GMM"),
  validate_cluster(rfm$Cluster_HClust, "Hierarchical Average"),
  validate_cluster(rfm$Cluster_PAM, "PAM / K-Medoids"),
  validate_cluster(rfm$Cluster_DBSCAN, "DBSCAN", exclude_noise = TRUE)
) |>
mutate(
  Rank.Silhouette = ifelse(is.na(Silhouette), Inf, min_rank(desc(Silhouette))),

```

```

Rank.Davies.Bouldin = ifelse(is.na(Davies.Bouldin), Inf, min_rank(Davies.Bouldin)),
Rank.Calinski.Harabasz = ifelse(is.na(Calinski.Harabasz), Inf, min_rank(desc(Calinski.Harabasz))),
Rank.Total = Rank.Silhouette + Rank.Davies.Bouldin + Rank.Calinski.Harabasz
) |>
arrange(Rank.Total, desc(Silhouette), Davies.Bouldin, desc(Calinski.Harabasz))

cluster_candidates <- list(
  "K-Means" = rfm$Cluster_KMeans,
  "GMM" = rfm$Cluster_GMM,
  "Hierarchical Average" = rfm$Cluster_HClust,
  "PAM / K-Medoids" = rfm$Cluster_PAM,
  "DBSCAN" = rfm$Cluster_DBSCAN
)

best_method <- validasi_semua$Metode[1]
rfm$Cluster_Final <- as.factor(cluster_candidates[[best_method]])

knitr::kable(validasi_semua,
  caption = "Rangkuman Validasi Internal dan Pemilihan Algoritma Terbaik")

```

Table 2: Rangkuman Validasi Internal dan Pemilihan Algoritma Terbaik

Metode	N	Jumlah Cluster	Noise	Silhouette	Stability	Davies.Bouldin	Calinski.Harabasz	Rank Silhouette	Rank Davies.Bouldin	Rank Calinski.Harabasz	Rank Total
K-Means	5878	2	0	0.4381	0.4381	0.9693	6205.27	2	2	1	5
Hierarchical Average	5878	2	0	0.5815	0.5815	0.7459	151.62	1	1	5	7
PAM / K-Medoids	5878	2	0	0.4340	0.4340	0.9765	6161.99	3	3	2	8
GMM	5878	2	0	0.2924	0.2924	1.0982	3056.20	4	4	3	11
DBSCAN	5819	2	59	0.2784	0.2784	1.1334	2902.03	5	5	4	14

```
cat("Algoritma clustering terbaik yang dipilih:", best_method, "\n")
```

```
## Algoritma clustering terbaik yang dipilih: K-Means
```

```
cat("Ukuran cluster final:\n")
```

```
## Ukuran cluster final:
```

```
print(table(rfm$Cluster_Final))
```

```
##
```

```
##      1      2
```

```
## 2357 3521
```

Interpretasi: Metode dengan Silhouette lebih tinggi, Calinski-Harabasz lebih besar, dan Davies Bouldin lebih kecil dianggap menghasilkan cluster yang lebih kompak dan terpisah dengan baik.

12.5 Tabel Kontingensi Antar-Algoritma

Tabel kontingensi digunakan untuk melihat kesesuaian hasil cluster antar metode. Pada bagian ini, hasil dari **algoritma terpilih** dibandingkan dengan algoritma lainnya. Semua algoritma diperlakukan sebagai kandidat yang setara.

```
for (metode in names(cluster_candidates)) {  
  if (metode != best_method) {  
    cat("\nAlgoritma terpilih vs", metode, ":\n")  
    print(table(Cluster_Final = rfm$Cluster_Final,  
                Pembanding    = as.factor(cluster_candidates[[metode]])))  
  }  
}
```

```
##
```

```
## Algoritma terpilih vs GMM :
```

```
##           Pembanding
```

```
## Cluster_Final      1      2
```

```
##           1      7 2350
```

```
##           2 1787 1734
```

```
##
```

```
## Algoritma terpilih vs Hierarchical Average :
```

```
##           Pembanding
```

```
## Cluster_Final      1      2
```

```
##           1     17 2340
```

```
##           2      0 3521
```

```
##
```

```
## Algoritma terpilih vs PAM / K-Medoids :
```

```
##           Pembanding
## Cluster_Final    1    2
##           1 2357    0
##           2  173 3348
##
## Algoritma terpilih vs DBSCAN :
##           Pembanding
## Cluster_Final    0    1    2
##           1  48 2309    0
##           2  11 1894 1616
```

13 Profil Cluster

Cluster yang digunakan untuk interpretasi adalah `Cluster_Final`, yaitu hasil dari algoritma terbaik berdasarkan tabel validasi internal semua algoritma.

```
profil <- rfm |>
  group_by(Cluster_Final) |>
  summarise(
    N          = n(),
    Recency    = round(mean(Recency), 1),
    Frequency  = round(mean(Frequency), 1),
    Monetary   = round(mean(Monetary), 0)
  )
knitr::kable(profil, align = "lrrrr",
  format.args = list(big.mark = ","),
  caption     = "Profil Rata-rata RFM per Cluster")
```

Table 3: Profil Rata-rata RFM per Cluster

Cluster_Final	N	Recency	Frequency	Monetary
1	2,357	51.4	12.6	6,604
2	3,521	302.6	2.1	619

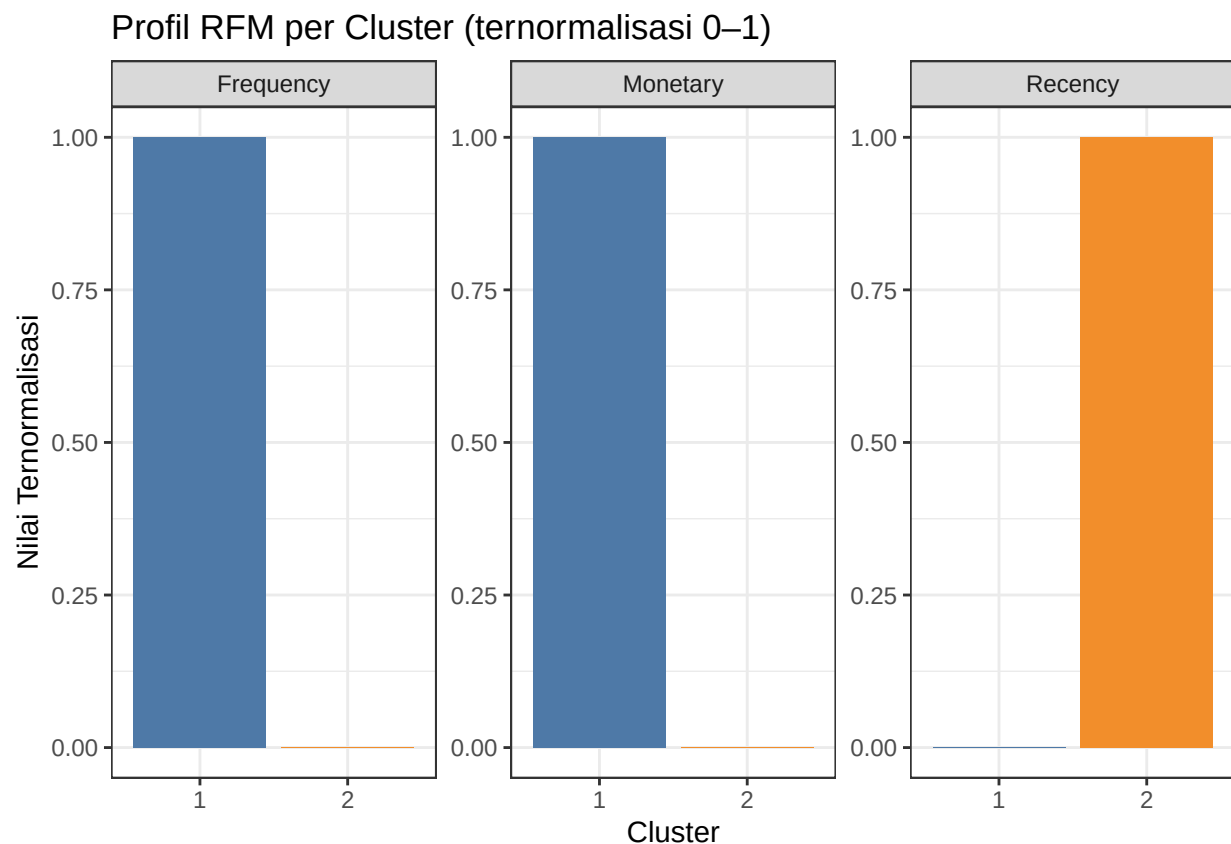
13.1 Visualisasi Profil Radar / Bar

```

profil_long <- profil |>
  tidyr::pivot_longer(cols = c(Recency, Frequency, Monetary),
                      names_to = "Metric",
                      values_to = "Value") |>
  # Normalisasi 0-1 per metrik untuk perbandingan
  group_by(Metric) |>
  mutate(ValueNorm = (Value - min(Value)) / (max(Value) - min(Value) + 1e-9)) |>
  ungroup()

ggplot(profil_long, aes(x = Cluster_Final, y = ValueNorm, fill = Cluster_Final)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ Metric, scales = "free_y") +
  scale_fill_manual(values = c("#4E79A7", "#F28E2B")) +
  labs(title = "Profil RFM per Cluster (ternormalisasi 0-1)",
       x = "Cluster",
       y = "Nilai Ternormalisasi") +
  theme_bw()

```



13.2 Interpretasi Setiap Cluster

Berikut interpretasi bisnis berdasarkan rata-rata RFM untuk 2 cluster berdasarkan output profil:

```
interp <- data.frame(  
  Cluster      = c("1","2"),  
  Segmen       = c("Pelanggan Aktif / Bernilai Tinggi",  
                   "Pelanggan Tidak Aktif / Bernilai Rendah"),  
  Deskripsi     = c(  
    "Recency rendah, Frequency tinggi, dan Monetary tinggi; pelanggan masih aktif serta member:  
    "Recency tinggi, Frequency rendah, dan Monetary rendah; pelanggan cenderung tidak aktif dan  
  ),  
  Rekomendasi  = c(  
    "Pertahankan dengan program loyalitas, rekomendasi personal, dan penawaran eksklusif.",  
    "Lakukan kampanye reaktivasi, diskon terbatas, dan reminder produk yang relevan."  
  ),  
  stringsAsFactors = FALSE  
)  
knitr::kable(interp, align = "l",  
              caption = "Interpretasi dan Rekomendasi Bisnis per Cluster")
```

Table 4: Interpretasi dan Rekomendasi Bisnis per Cluster

Cluster	Segmen	Deskripsi	Rekomendasi
1	Pelanggan Aktif / Bernilai Tinggi	Recency rendah, Frequency tinggi, dan Monetary tinggi; pelanggan masih aktif serta memberikan nilai transaksi besar.	Pertahankan dengan program loyalitas, rekomendasi personal, dan penawaran eksklusif.
2	Pelanggan Tidak Aktif / Bernilai Rendah	Recency tinggi, Frequency rendah, dan Monetary rendah; pelanggan cenderung tidak aktif dan kontribusi transaksi lebih kecil.	Lakukan kampanye reaktivasi, diskon terbatas, dan reminder produk yang relevan.

Catatan: Label segmen di atas bersifat indikatif. Sesuaikan dengan nilai aktual pada tabel profil yang muncul saat knitr.